

Möbius Transformations

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In olympiad geometry, two important techniques that often appear are *inversive* and *projective* geometry. Some of you may know that these two realms of geometry interact nicely in many ways. For example, the cross-ratio, the most important quantity in olympiad projective geometry, is conjugated by inversion. Today we'll delve into this relationship deeper by discussing the Riemann sphere $\widehat{\mathbb{C}}$ and its properties. Notably we'll discuss the automorphisms of $\widehat{\mathbb{C}}$, known as the *Möbius transformations*, and develop an understanding of how they operate.

In this handout I take several figures from [Nee23]. All credit for these figures belongs to Needham.

1 Inversion Review

Throughout, given a map f and a geometric object Γ , we say that f *preserves* Γ if $f(\Gamma) = \Gamma$. Notably f preserving Γ will not necessarily mean that f fixes every point on Γ .

Definition 1.1. Given a circle Ω with center O and radius r , *inversion* about Ω is the operation which takes a point P to the point Q on ray $\overrightarrow{OP}$ satisfying $OQ \times OP = r^2$.

This technically does not define where O goes to. But we can view it as going to the “point at infinity”, and we will make this rigorous soon.

Theorem 1.2. Consider a circle Ω with center O , and let ϕ be the operation given by inverting about Ω . Then ϕ has the following properties:

1. ϕ is an involution, i.e. $\phi \circ \phi = \text{id}$.
2. ϕ preserves lines passing through O .
3. ϕ maps lines not passing through O to circles passing through O , and vice versa.
4. ϕ maps circles not passing through O to other circles not passing through O .

Proof. Angle chasing. □

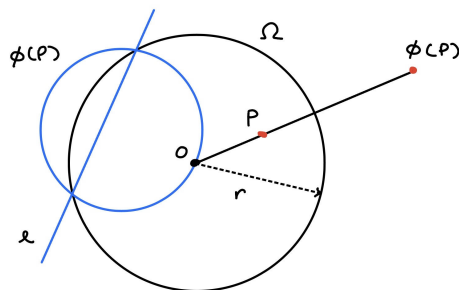


Figure 1: Here ϕ denotes inversion about the circle Ω .

Example 1.3. Let Ω be the unit circle in the complex plane. Then inversion about Ω is given by

$$z \mapsto \frac{1}{\bar{z}}.$$

In this sense we may view inversion as an antiholomorphic function on the *Riemann sphere* $\hat{\mathbb{C}} = \mathbb{C} \cup \{\infty\}$.

We can deduce many important properties about inversion using the above properties, and the fact that they are bijections (in fact involutions). For example, generally define a *cline* as a circle or a line, where lines are viewed as circles which pass through $\infty \in \hat{\mathbb{C}}$. Then inversions must preserve the number of intersections between any two clines, so two tangent clines remain tangent under inversion. Notably, two *lines* in $\hat{\mathbb{C}}$ are tangent iff they are parallel, so inverting two tangent circles about a circle centered at their point of tangency gives two parallel lines.

Going off of this idea, inversion is generally useful to try when you have many circles passing through a point.

Example 1.4. [Ukraine NMO 2025 10.7] Given a triangle ABC , an arbitrary point D is chosen on the side AC . In triangles ABD and CBD , the angle bisectors BK and BL are drawn, respectively. The point O is the circumcenter of $\triangle KBL$. Prove that the second intersection point of the circumcircles of triangles ABL and CBK lies on the line OD .

1. Perform an inversion about the point B . (You can make this a $\sqrt{ac}$ inversion if you wish).
2. Let X be the reflection of B across O ; then BX is a diameter of (BKL) . Using this, characterize the image of X , hence also O , under the inversion.
3. Finish the problem.

Remark 1.5. You can deduce most properties you'll ever need about an inverted diagram using Theorem 1.2. Thus, even if inversion feels like a rather complicated technique to you now, a good knowledge of Theorem 1.2 should make inversion a relatively lower-cost technique for olympiad geometry standards.

1.1 Orthogonality

When is a circle ω fixed under inversion about another circle Ω ?

Definition 1.6. Two circles Ω_1 and Ω_2 with centers O_1, O_2 (respectively) are *orthogonal* if the intersections P, Q of Ω_1 and Ω_2 satisfy $\angle O_1PO_2 = \angle O_1QO_2 = 90^\circ$.

Example 1.7. Prove that inversion about Ω fixes ω iff Ω and ω are orthogonal.

Orthogonality is a particularly useful condition because it is *reflexive*. Notably given two circles Ω, ω it implies that the inversion about Ω fixes ω iff the inversion about ω fixes Ω ! It also shows up in a lot of contexts, such as the following.

Example 1.8. Consider a circle Ω and a point P outside of Ω . Let the tangents from P to Ω touch Ω at R and S . Given any other point Q in the plane, prove that the circle with diameter PQ is orthogonal to Ω iff $Q \in RS$.

For example, this solves the Warmup G3 from earlier:

Example 1.9. Consider three circles $\Gamma_1, \Gamma_2, \Gamma_3$ which are non-overlapping and mutually external. For each point P in the plane outside the three circles, construct six points $A_1, B_1, A_2, B_2, A_3, B_3$ as follows: for each $i = 1, 2, 3$, A_i, B_i are distinct points on the circle Γ_i such that the lines PA_i and PB_i are both tangents to Γ_i . Call a point P exceptional if, from the construction, the three lines A_1B_1, A_2B_2, A_3B_3 are concurrent. Show that every exceptional point of the plane lies on a same circle.

2 Cross-Ratio Recap

Definition 2.1. Given four (distinct) complex points $z_1, z_2, z_3, z_4 \in \mathbb{C}$, their *cross-ratio* is defined as

$$[z_1, z_2; z_3, z_4] := \frac{z_1 - z_3}{z_1 - z_4} \div \frac{z_2 - z_3}{z_2 - z_4}.$$

Note that the cross ratio being real is equivalent to z_1, z_2, z_3, z_4 being concyclic (or collinear, if one of z_1, z_2, z_3, z_4 is ∞).

Note that cross ratios satisfy that

$$[z_1, z_2; z_3, z_4] = [z_2, z_1; z_3, z_4]^{-1} = [z_2, z_1; z_4, z_3] = [z_3, z_4; z_1, z_2].$$

Cross ratios can also be defined for lines:

Definition 2.2. Consider five distinct points A, B, C, D, P , where A, B, C, D are collinear. Then define the cross ratio $[PA, PB; PC, PD]$ as

$$[PA, PB; PC, PD] := [A, B; C, D] = \frac{\sin(\angle APC)}{\sin(\angle APD)} \div \frac{\sin(\angle BPC)}{\sin(\angle BPD)}.$$

Note that the second equality implies that the cross ratio $[PA, PB; PC, PD]$ is independent of the choice of A, B, C, D , and only depends on the lines PA, PB, PC, PD themselves. Using this, one can show:

Lemma 2.3. Fix two lines l, l' . Consider four other lines l_1, l_2, l_3, l_4 , all passing through a point P . Denote $A_i := l \cap l_i$, $A'_i := l' \cap l_i$. Then

$$[A_1, A_2; A_3, A_4] = [A'_1, A'_2; A'_3, A'_4].$$

From your earlier olympiad training, you may recall many useful facts about a 4-tuple which is *harmonic*, i.e. has cross ratio is -1 . For instance, given A, B, C, D collinear, and a point P satisfying $\angle APC = 90^\circ$, $[A, B; C, D] = -1$ is equivalent to $\angle BPC = \angle DPC$. We will not rehash these properties here, but note that even for non-harmonic quadrilaterals, the cross-ratio can be a useful computational tool, since for fixed z_2, z_3, z_4 , the map

$$z_1 \rightarrow [z_1, z_2; z_3, z_4]$$

is injective. For instance, we can use this injectivity as follows.

Lemma 2.4 (Prism Lemma). Consider two lines l, l' and points $A, B, C, D \in l, A', B', C', D' \in l'$, such that AA', BB', CC' concur at a point P . Then DD' also passes through P iff

$$[A, B; C, D] = [A', B'; C', D'].$$

Proof. Suppose PD intersects l' at D'' . Then $[A', B'; C', D''] = [A, B; C, D]$, so $D' = D''$ iff $[A', B'; C', D'] = [A, B; C, D]$. $\square$

This can be a very useful tool for translating collinearity problems into length computations. For instance, a few days ago some of you were tasked with the following problem by Kate:

Example 2.5. Let ABC be a triangle with $\angle A = 90^\circ$. Let I be the incenter, E and F the B and C -intouch points, and Q and P the B and C -extouch points (respectively). Let $M \in EF$ such that MI is parallel to BC . Prove that M, P, Q are collinear.

This problem is pretty hard to directly do synthetically (see #random in the discord for a solution). But we can make this amenable to computation, by noting that if $MI \cap AB = X$, $MI \cap AC = Y$, then the problem is equivalent to showing $(A, X; F, P) = (A, Y; E, Q)$, and all the relevant lengths are easy to compute.

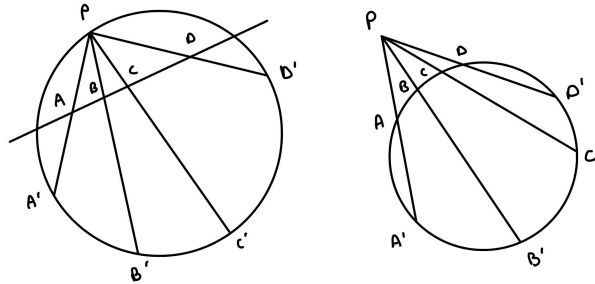
2.1 Inversive Relations

It turns out that inversions and cross-ratios interact very nicely!

Lemma 2.6. *Cross-ratios are conjugated under inversion. That is, consider points P, A, B, C, D , and let A', B', C', D' be the images of A, B, C, D under inversion about P by arbitrary radius. Then*

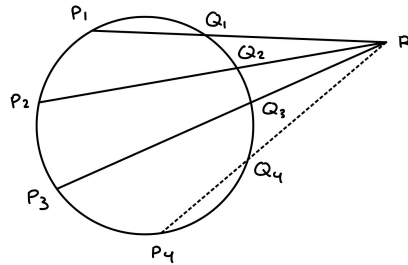
$$[A, B; C, D] = \overline{[A', B'; C', D']}.$$

Example 2.7. Prove that $[A, B; C, D] = [A', B'; C', D']$ in the examples below.



Notably, we can use this to setup a [cyclic prism lemma](#).

Lemma 2.8 (Cyclic Prism Lemma). *Consider a circle Ω and points $\{P_i, Q_i : i \in [1, 4]\}$ such that P_1Q_1, P_2Q_2, P_3Q_3 concur at a point R . Then $R \in P_4Q_4$ iff $[P_1, P_2; P_3, P_4] = [Q_1, Q_2; Q_3, Q_4]$.*



Example 2.9. Prove the Cyclic Prism Lemma.

3 Möbius Transformations

The fact that inversions conjugate cross-ratios is equivalent to the inversion-reflection $z \mapsto 1/\bar{z}$ preserving the cross-ratio. This fits into the following more general theory.

Definition 3.1. A [Möbius transformation](#) is a map $f : \widehat{\mathbb{C}} \rightarrow \widehat{\mathbb{C}}$ given by

$$f(z) := \frac{az + b}{cz + d}$$

for some $a, b, c, d \in \mathbb{C}$, such that (a, b) and (c, d) are not scalar multiples of each other (so $f(z)$ is not a constant). Here we define $f(\infty) = \lim_{z \rightarrow \infty} f(z)$, and if $cz + d = 0$ then $f(z) = \infty$.

Example 3.2. Prove that the Möbius transformations form a group under composition. We will denote this group [Aut \$\widehat{\mathbb{C}}\$](#) ; this suggestive notation is to illustrate that the Möbius transformations are precisely the *automorphisms* of $\widehat{\mathbb{C}}$ as a variety / complex manifold, though you do not need to know what this means.

Möbius transformations are all compositions of *rotations*, *translations*, and *reflection-inversions*. We need our inversions to be reflected since $z \mapsto 1/\bar{z}$ is *not* a Möbius transformation while $z \mapsto 1/z$ *is*. Anyways this distinction does not matter too much, and if don't want to reflect then you're free to complex conjugate your f .

Möbius transformations are particularly useful since they satisfy several *transitivity properties*. The most important one is the following.

Lemma 3.3. *Möbius transformations are simply triply transitive; that is, a Möbius transformation is determined by where it sends any three distinct points in $\widehat{\mathbb{C}}$.*

Proof. Uniqueness is by cross-multiplying an equation $f(z) = g(z)$ and noting both sides form quadratics. For existence, since $\text{Aut } \widehat{\mathbb{C}}$ is a group, it suffices to construct a Möbius transformation mapping $\{0, 1, \infty\}$ to any ordered triplet of distinct points in $\widehat{\mathbb{C}}$, which is a direct computation. $\square$

Example 3.4. Find a Möbius transformation which sends the open unit disk in $\mathbb{C}$ to the upper half plane $\mathbb{H} = \{z : \text{Im}(z) > 0\}$. Where does your Möbius transformation send the points $\{-1, 0, 1\}$? Is your choice of Möbius transformation unique?

Using this simple transitivity property, we can classify Möbius transformations as follows. Observe that by cross-multiplying the equation $f(z) = \frac{az+b}{cz+d} = z$, we know that whenever $f(z) \neq z$ identically, $f(z)$ has at most 2 fixed points. Now:

1. If $f(z)$ has two fixed points $\{p, q\}$, pick some $g \in \text{Aut } \widehat{\mathbb{C}}$ mapping $g(0) = p$, $g(\infty) = q$. Then $h := g^{-1} \circ f \circ g$ is a Möbius transformation with fixed points $\{0, \infty\}$, so by direct computation it must be of the form $z \mapsto az$ for $a \in \mathbb{C}^\times$. So h is a composition of a rotation and dilation about the origin, and we can understand f as a conjugate of h . Generally we will call f *loxodromic*; if $h = g^{-1}fg = \{z \mapsto az\}$ satisfies $a \in S^1$ (so f is conjugate to a rotation) then we call f *elliptic*, while if $a \in \mathbb{R}^\times$ (so f is conjugate to a dilation) then we call f *hyperbolic*. See Figures 3, 4, and 5 at the end of this handout for some pictures.
2. If $f(z)$ has one fixed point $\{p\}$, then pick some $g \in \text{Aut } \widehat{\mathbb{C}}$ such that $g(\infty) = p$. Then $h = g^{-1} \circ f \circ g$ is a Möbius transformation whose only fixed point is ∞ , so it must be a translation of the form $z + c$ for $c \in \mathbb{C}$. In this case we call f *parabolic*.

Example 3.5. Determine if the Möbius transformation $z \mapsto 1/z$ is loxodromic or parabolic. If it is loxodromic, determine if it is elliptic, hyperbolic, or neither.

3.1 Geometric Properties

We can generalize many properties about inversion naturally to the setting of Möbius transformations. For instance, since inversions, reflections, rotations, dilations, and translations all preserve the cross-ratio, all Möbius transformations should. Indeed, we may directly compute that for all $f \in \text{Aut } \widehat{\mathbb{C}}$,

$$[f(z_1), f(z_2); f(z_3), f(z_4)] = [z_1, z_2; z_3, z_4].$$

Notably, observe that four points z_1, z_2, z_3, z_4 have real cross-ratio iff they all lie on a cline. Thus Möbius transformations also send clines to clines.

Möbius transformations also have the property that they are *conformal*, i.e. they preserve local angle. This is because they are everywhere complex-differentiable (where to define this notion at ∞ , work under the chart $z \mapsto 1/z$; ignore this if this sounds like nonsense to you), so they act locally by a complex rotation (their derivative in $\mathbb{C}$) at every point. One major consequence of this is that Möbius transformations preserve *orthogonality of clines*, since e.g. two circles are orthogonal iff at any of their intersections, the two tangents to the arcs are orthogonal.

Example 3.6. Let $\mathbb{H} = \{z : \text{Im}(z) > 0\}$ be the upper half-plane, and $\Delta = \{z : |z| < 1\}$ be the open unit disk. Earlier we constructed a Möbius transformation $g \in \text{Aut } \widehat{\mathbb{C}}$ taking Δ to $\mathbb{H}$. Now, let $\text{Aut } \mathbb{H} \subset \text{Aut } \widehat{\mathbb{C}}$ be the subgroup of Möbius transformations mapping $\mathbb{H}$ to itself, and define $\text{Aut } \Delta$ similarly.

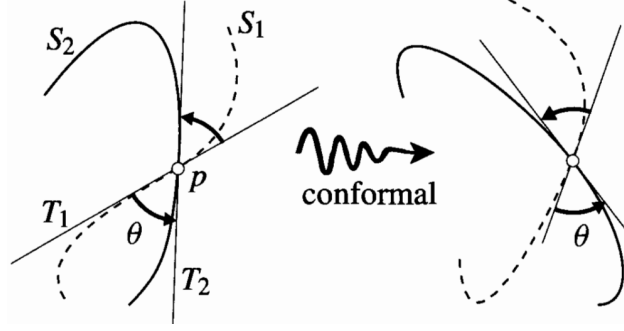


Figure 2: Taken from [Nee23, Figure 8]. A diagram showing how conformal maps act locally.

1. Prove that

$$\text{Aut } \mathbb{H} = \left\{ f(z) = \frac{az + b}{cz + d} : a, b, c, d \in \mathbb{R}, ad - bc > 0 \right\}.$$

By conjugating by g , also derive an explicit form for $\text{Aut } \Delta$.

2. Prove that given any circle $\omega \subset S^1$, there exists some $f \in \text{Aut } \Delta$ mapping ω to a radial circle rS^1 (for some r), as follows. Pick a diameter PQ of S^1 orthogonal to ω (i.e. the center of ω lies on PQ). Suppose $\omega \cap PQ = \{P_1, Q_1\}$. Then pick points $P_2, Q_2 \in PQ$, such that P_2 and Q_2 are reflections about the origin, and

$$(P, Q; P_2, Q_2) = (P, Q; P_1, Q_1).$$

Then construct f using P_2 and Q_2 .

3.2 Steiner Chains

Here's a really powerful application of the above transitivity results.

Definition 3.7. Given circles Ω, Γ such that Ω is contained in the interior disk bounded by Γ , a *Steiner chain of length n* is a sequence of n circles $\omega_1, \dots, \omega_n$, such that for each i , ω_i is externally tangent to Ω , ω_{i-1} , and ω_{i+1} , and internally tangent to Γ .

Example 3.8. Consider a Steiner chain $\omega_1, \dots, \omega_{2n}$ of length $2n$ for two circles Ω, Γ . Say ω_i is tangent to ω_{i+1} at T_i .

1. Prove that the points T_i all lie on a circle.
2. Prove the lines $T_i T_{i+n}$ all concur.

It turns out that this problem is almost immediate from the ideas above.

1. Conjugate by a Möbius transformation which maps (Γ, Ω) to a pair of concentric circles. Why does this solve 1)?
2. For 2), consider where ∞ maps under the transformation above.

Remark 3.9 (Remark for Experts). One technical note: today we discussed there being a “complex point at infinity”, but in projective geometry you are more accustomed to there being a “line at infinity”. The reason for this discrepancy is as follows. Usually, when we work with projective geometry, we are working in the *real projective plane* $\mathbb{RP}^2$. This has a real “projective line at infinity”, whose complement is $\mathbb{R}^2$. Today, instead, we worked with the *complex projective line* $\mathbb{CP}^1$, which has a complex point at infinity whose complement is $\mathbb{C}$. This is also why we could take the cross-ratio of any 4 points: you can always take the cross-ratio of 4 *complex-collinear* points, but this is true for any 4 points in $\mathbb{CP}^1$ by definition.

4 Problems

And Möbius said
 “It’s Mobin Time!”
 and Mobed all over the diagram.

August Ferdinand Möbius, probably

The contest problems are mostly inversion and projective practice. However the Theoretical practice will get you very familiar with different constructions about inversion and Möbius transformations.

4.1 Theroetical Practice

Every problem in this section is useful and should be ★ starred.

1. Let a *quasi-harmonic* quadrilateral be a quadrilateral $ABCD$ satisfying $AB \cdot CD = AD \cdot BC$. What happens if you invert a quasi-harmonic quadrilateral about A ?
2. Consider two circles Ω_1, Ω_2 intersecting at points P and Q , and suppose their external tangents intersect at a point X . Prove that inversion about the circle (X, XP) swaps Ω_1 and Ω_2 .
3. Consider two circles Ω and ω , and suppose Ω has center O . A line through O intersects ω at P and Q . Prove that ω and Ω are orthogonal iff inversion about Ω swaps P and Q .
4. If $(A, B; C, D) = -1$, prove that any circle passing through C and D is orthogonal to the circle with diameter (AB) .
5. Given a point O and a real $r > 0$, *negative inversion* about O with “radius” r maps any point P to the point Q such that $OP \cdot OQ = r^2$, and O lies on segment PQ . In other words, you invert about the circle centered at O with radius r , then reflect across O .
 - (a) How do you express negative inversions in terms of complex numbers?
 - (b) Consider a triangle ABC with orthocenter H . Let $AH \cap BC = D, BH \cap CA = E, CH \cap AB = F$. Can you find a negative inversion about H which swaps $\{A, B, C\}$ and $\{D, E, F\}$? Where does this map the circle (ABC) ?
6. Given a triangle ABC , $\sqrt{bc}$ *inversion* is the operation which first inverts about the circle centered at A with radius $\sqrt{AB \cdot AC}$, then reflects across the angle bisector of $\angle BAC$.
 - (a) Express $\sqrt{bc}$ inversion in terms of complex numbers.
 - (b) Given any point $D \neq A$ in the plane, let E be its image under $\sqrt{bc}$ inversion. Prove $ADB \sim ACE$.
7. We can in fact view cross-ratios as Möbius transformations themselves!
 - (a) Given fixed distinct $z_2, z_3, z_4 \in \widehat{\mathbb{C}}$, prove that $z_1 \mapsto [z_1, z_2; z_3, z_4]$ is a Möbius transformation.
 - (b) Using this construction, prove that the group of Möbius transformations $\text{Aut } \widehat{\mathbb{C}}$ is simply triply transitive.
8. Consider a circle Ω . Given any two distinct points $P, Q \in \Omega$ such that PQ is not a diameter of Ω , let ω_{PQ} be the unique circle passing through P and Q orthogonal to Ω . Given four distinct points $A, B, C, D \in \Omega$ lying in that order, prove there exists a circle ω externally tangent to all the circles $\omega_{AB}, \omega_{BC}, \omega_{CD}, \omega_{DA}$ if and only iff $AB \times CD = AD \times BC$.
9. Let $f(z) = \frac{az+b}{cz+d}$ be a Möbius transformation. Show that the number of rational maps $g : \widehat{\mathbb{C}} \rightarrow \widehat{\mathbb{C}}$ such that

$$g(g(g(g(g(z)))))) = f(z)$$

is 1, 5, or ∞ . Explain how to determine which alternative holds for a given f .

4.2 Contest Practice

4.3 Part A

Here are some IMO 1 level problems.

- Two circles intersect at two points A and B . A line l passing through A meets these circles again at C and D , such that C, D lie on the arcs AB of the two circles which are not contained in the other circle. Let M, N respectively be the midpoints of the arcs BC, BD (not containing A) of their respective circles. Let K be the midpoint of segment CD . Prove $\angle MKN = 90^\circ$.
- ★ Consider a regular hexagon $A_1A_2A_3A_4A_5A_6$, and a point P in the plane which is not on any of A_1A_4, A_2A_5, A_3A_6 . Prove that the circles $(PA_1A_4), (PA_2A_5), (PA_3A_6)$ concur at a point distinct from P .
- Let ABC be a triangle with circumcircle Ω and A -excenter J . The incircle of triangle ABC touches its side BC at D . Let K, L be the two intersection points of Ω and the circle with diameter DJ , and lines DK and DL meet Ω again at P and Q , respectively. Prove that line PQ passes through the incenter of triangle ABC .
- In a convex quadrilateral $ABCD$, $\angle ABC = \angle CDA$. Let $X \neq C$ be the intersection of the circumcircle of $\triangle BCD$ and circle with diameter AC . Prove that the tangent to the circumcircle of $\triangle BCD$ at X , the tangent to the circumcircle of $\triangle ABD$ at A concur on BD .
- Let ω_1 and ω_2 be two orthogonal circles, and let the center of ω_1 be O . Diameter AB of ω_1 is selected so that B lies strictly inside ω_2 . The two circles tangent to ω_2 , passing through O and A , touch ω_2 at F and G . Prove that $FGOB$ is cyclic.
- ★ Let A, B, C, D be four points in the plane, with C and D on the same side of the line AB , such that $AC \cdot BD = AD \cdot BC$ and $\angle ADB = 90^\circ + \angle ACB$. Find the ratio

$$\frac{AB \cdot CD}{AC \cdot BD},$$

and prove that the circumcircles of the triangles ACD and BCD are orthogonal.

4.4 Part B

Here are some IMO 2 level problems.

- ★ Consider three mutually intersecting circles $\Omega_1, \Omega_2, \Omega_3$. Suppose Ω_i and Ω_{i+1} share a common chord l_{i+2} , and these three common chords intersect at a point P . Also suppose there exists two circles ω, Ω internally tangent to all three circles.
 - Prove that P is either the exsimilicenter or insimilicenter of ω and Ω .
 - Which center is it?
- Let $ABCD$ be a cyclic quadrilateral with circumcenter O . Let P be the intersection of AC and BD . Points X and Y are on the plane such that $PX \parallel AB$ and $PY \parallel CD$. It is also known that AC bisects $\angle XCD$ and BD bisects $\angle ABY$. Prove that $XY \perp OP$.
- In the isosceles triangle ABC with $AB = AC$, the center of $\odot O$ is the midpoint of the side BC , and AB, AC are tangent to the circle at points E, F respectively. Point G is on $\odot O$ with $\angle AGE = 90^\circ$. A tangent line of $\odot O$ passes through G , and meets AC at K . Prove that line BK bisects EF .
- Let ABC be a triangle with $\angle C = 90^\circ$, and let H be the foot of the altitude from C . A point D is chosen inside the triangle CBH so that CH bisects AD . Let P be the intersection point of the lines BD and CH . Let ω be the semicircle with diameter BD that meets the segment CB at an interior point. A line through P is tangent to ω at Q . Prove that the lines CQ and AD meet on ω .

5. Let $ABCD$ be a convex quadrilateral and let P and Q be points in $ABCD$ such that $PQDA$ and $QPBC$ are cyclic quadrilaterals. Suppose that there exists a point E on the line segment PQ such that $\angle PAE = \angle QDE$ and $\angle PBE = \angle QCE$. Show that the quadrilateral $ABCD$ is cyclic.
6. ★ Consider a Steiner chain $(\omega_1, \dots, \omega_n)$ for two circles Ω, Γ , and as before say ω_i is tangent to ω_{i+1} at T_i . Recall that the points T_i all lie on a circle ω .
 - (a) Prove that for any $k \in [n]$, the lines $T_i T_{k-i}$ concur, where our indices are taken cyclically.
 - (b) Can you prove the above without using Möbius transformations?
7. ★ Five points A, B, C, D and E lie on a circle τ clockwise in that order such that $AB \parallel CE$ and $\angle ABC > 90^\circ$. Let k be a circle tangent to AD, CE and τ such that k and τ touch on the arc $\widehat{DE}$ not containing A, B and C . Let $F \neq A$ be the intersection of τ and the tangent line to k passing through A different from AD .
Prove that there exists a circle tangent to BD, BF, CE and τ .
8. Let Γ be a circle of radius 1 centered at O . A circle Ω is said to be *friendly* if there exist distinct circles $\omega_1, \omega_2, \dots, \omega_{2020}$, such that for all $1 \leq i \leq 2020$, ω_i is tangent to Γ, Ω , and ω_{i+1} . (Here, $\omega_{2021} = \omega_1$.) For each point P in the plane, let $f(P)$ denote the sum of the areas of all friendly circles centered at P . If A and B are points such that $OA = \frac{1}{2}$ and $OB = \frac{1}{3}$, determine $f(A) - f(B)$.
9. Let ABC be an acute scalene triangle with circumcenter O , and let T be on line BC such that $\angle TAO = 90^\circ$. The circle with diameter $\overline{AT}$ intersects the circumcircle of $\triangle BOC$ at two points A_1 and A_2 , where $OA_1 < OA_2$. Points B_1, B_2, C_1, C_2 are defined analogously. Prove that $\overline{AA_1}, \overline{BB_1}, \overline{CC_1}$ are concurrent. Prove that $\overline{AA_2}, \overline{BB_2}, \overline{CC_2}$ are concurrent on the Euler line of triangle ABC .
10. ★ Triangle ABC has incircle ω and A -excircle ω_A . Circle γ_B passes through B and is externally tangent to ω and ω_A . Circle γ_C passes through C and is externally tangent to ω and ω_A . If γ_B intersects line BC again at D , and γ_C intersects line BC again at E , prove that $BD = EC$.

4.5 Part C

Here are some IMO 3 level problems. Note the large difficulty jump from Part B to Part C.

1. ★ Let I be the incentre of a non-equilateral triangle ABC , I_A be the A -excentre, I'_A be the reflection of I_A in BC , and l_A be the reflection of line AI'_A in AI . Define points I_B, I'_B and line l_B analogously. Let P be the intersection point of l_A and l_B .
 - (a) Prove that P lies on line OI where O is the circumcentre of triangle ABC .
 - (b) Let one of the tangents from P to the incircle of triangle ABC meet the circumcircle at points X and Y . Show that $\angle XIY = 120^\circ$.
2. Let ABC be a triangle with $CA \neq CB$. Let D, F , and G be the midpoints of the sides AB, AC , and BC respectively. A circle Γ passing through C and tangent to AB at D meets the segments AF and BG at H and I , respectively. The points H' and I' are symmetric to H and I about F and G , respectively. The line $H'I'$ meets CD and FG at Q and M , respectively. The line CM meets Γ again at P . Prove that $CQ = QP$.
3. ★ A convex quadrilateral $ABCD$ satisfies $AB \cdot CD = BC \cdot DA$. Point X lies inside $ABCD$ so that

$$\angle XAB = \angle XCD \quad \text{and} \quad \angle XBC = \angle XDA.$$

Prove that $\angle BXA + \angle DXC = 180^\circ$.

References

[Nee23] Tristan Needham. *Visual complex analysis*. Oxford University Press, 2023.

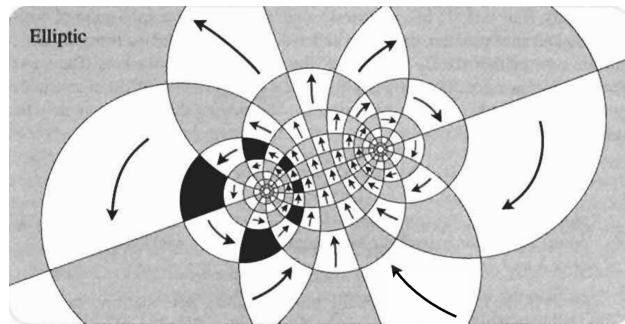


Figure 3: [Nee23, Figure 30]. A picture of an elliptic Möbius transformation

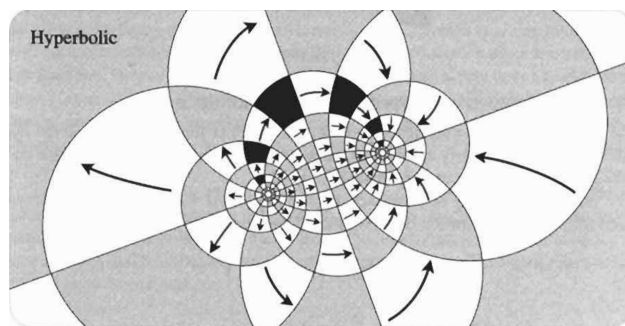


Figure 4: [Nee23, Figure 31]. A picture of a hyperbolic Möbius transformation

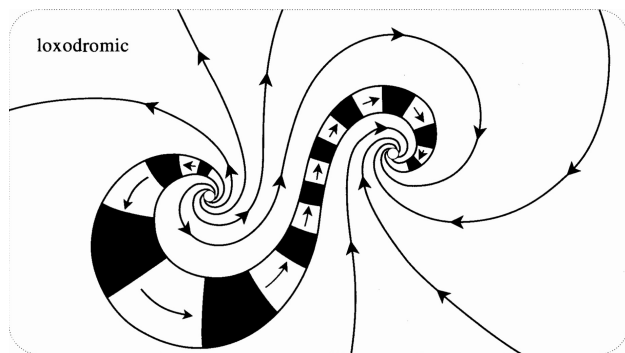


Figure 5: [Nee23, Figure 32]. A picture of a loxodromic Möbius transformation